From Biochemical Hits to Pharmacologically Credible Leads: A Critical Appraisal of Heparanase Inhibitor Discovery

Abstract

Heparanase (HPSE1) remodels heparan sulfate in extracellular and pericellular environments and contributes to processes associated with malignancy, inflammation, and vascular injury. Numerous inhibitors show biochemical activity, yet progression toward clinically useful drugs has been limited. This critical narrative review evaluates how evidence for representative inhibitor classes supports pharmacologically credible claims, integrating experimentally supported structure–activity relationships (SAR) with assay context, mechanism, selectivity, target engagement, exposure, and clinical outcomes. The analysis finds that potency depends strongly on substrate and assay architecture; selectivity and cellular target engagement remain unevenly characterized; and preclinical efficacy does not by itself establish an HPSE-specific mechanism. Clinical findings reinforce this distinction: PI-88 did not meet the primary disease-free-survival endpoint in its Phase III PATRON trial, while partial responses with pixatimod were reported in a Phase Ib nivolumab combination and cannot be attributed to HPSE inhibition alone. Matched structural changes can yield context-dependent, even reversed, effects, so inhibitor classes provide complementary evidence profiles rather than a universal potency hierarchy. Future programs should prioritize orthogonal assays, modality-appropriate selectivity, and exposure-linked pharmacodynamics.

Keywords: Heparanase; HPSE1; heparan sulfate; inhibitor discovery; medicinal chemistry; structure–activity relationship; selectivity; target engagement; pharmacokinetics; translational pharmacology.

1 Introduction

Heparan sulfate proteoglycans connect extracellular matrix organization to growth-factor signaling, inflammatory-cell movement, tissue repair, and vascular barrier function. Their heparan sulfate chains organize interactions with proteins rather than acting only as passive structural components. Heparanase-1 (HPSE1), classically described as a mammalian endo-β-D-glucuronidase, cleaves these chains and can alter both the physical integrity of tissue barriers and the availability of heparan sulfate-associated signaling molecules. Structural studies established a catalytic machinery centered on Glu225 and Glu343, whereas recent experiments with defined substrates suggest that cleavage behavior can depend on substrate sequence and context [1, 26, 37].

Abnormal heparanase expression or activity has been associated with tumor invasion, angiogenesis, metastatic dissemination, treatment resistance, inflammation, and microvascular dysfunction. These associations have motivated inhibitor programs using sulfated polysaccharides, oligosaccharides, synthetic

glycomimetics, heterocyclic small molecules, irreversible glycosidase-directed inhibitors, and more recently induced protein degradation. The mechanistic breadth of heparanase biology creates opportunity but complicates interpretation. Reduced cellular invasion after compound exposure could reflect direct catalytic inhibition, disruption of growth-factor binding, altered receptor signaling, nonspecific charge-dependent interactions, or combinations of these mechanisms [3, 5, 44].

The historical emphasis on enzyme potency has produced a diverse chemical literature, while clinically tested agents have largely been heparan sulfate mimetics. Zhang and Cui [2] reviewed small-molecule discovery and screening. This review connects experimentally supported within-series SAR with assay dependence, binding mechanism, modality-matched selectivity, target engagement, exposure, and clinical outcomes. Small molecules are the primary focus; chemically defined and clinically investigated glycomimetics provide comparators across these evidence domains.

This narrative review integrates medicinal chemistry, biochemical, biophysical, pharmacological, and clinical literature. Focused searches of PubMed, Scopus, ScienceDirect, and Google Scholar combined terms for heparanase or HPSE with inhibitor discovery, heparan sulfate mimetics, structure–activity relationships, fragment screening, and structure-based design. The review emphasizes literature published from 2020 through 2026, alongside earlier studies establishing structural, chemical, and clinical foundations. Studies were selected qualitatively for evidence on inhibitor chemistry, activity, mechanism, selectivity, pharmacology, or clinical outcomes. The literature search was last updated on 25 October 2026.

2 Structural and biochemical context for inhibitor design

2.1 Catalytic site and substrate recognition

Active human HPSE1 is generated through proteolytic maturation of a precursor and contains associated subunits forming the catalytic architecture. Glu225 acts as the catalytic acid/base, and Glu343 participates as the nucleophile in glycosidic bond cleavage. The groove surrounding the active site accommodates sulfated carbohydrate motifs through a network of electrostatic, polar, and hydrophobic contacts. The crystal structures analyzed by Zhang et al. [26] include enzyme complexes with a heparin-derived tetrasaccharide and with small molecules, providing a basis for discussing how a candidate ligand may recognize either catalytic or adjacent substrate-binding regions.

Substrate recognition is not governed by negative charge alone. Sulfation position, uronic-acid configuration, chain length, conformational presentation, and accessibility can affect substrate turnover and ligand recognition. Wang et al. [37], using structurally defined oligosaccharides, reported substrate-dependent cleavage behavior that complicates an exclusively endo-acting description of the enzyme. This result is particularly relevant to inhibitor studies: assays using short synthetic substrates, fondaparinux-related fragments, fluorescent probes, or matrix-embedded heparan sulfate are not necessarily reporting inhibition under equivalent molecular conditions.

2.2 Beyond a single catalytic pocket

A compound detected in the catalytic cleft may inhibit by blocking substrate access or engaging residues required for catalysis, but crystallographic binding alone does not establish functional inhibition. Davies et al. [25] identified 31 structurally observed fragment-binding events, including orthosteric and allosteric sites. This establishes experimentally accessible binding opportunities beyond the conventional substrate-mimetic pocket; the degree of enzyme inhibition and optimization is considered with the medicinal chemistry evidence in Section 3.2.5.

Whitefield et al. [28] found that pentosan polysulfate inhibition involved multiple overlapping binding events affected by oligomer length and protein structural changes. These results contrast with a simple one-site competitive model and underscore the mechanistic complexity of this glycomimetic.

3 Inhibitor chemotypes and evidence quality

3.1 Heparan sulfate mimetics as development benchmarks

3.1.1 Clinically investigated sulfated agents

Molecularly complex and highly sulfated agents provided some of the earliest advanced pharmacological tests of the HPSE hypothesis. Muparfostat (PI-88), roneparstat (SST0001), pixatimod (PG545), and necuparanib (M-402) entered clinical development in different disease settings. Their chain length, sulfation pattern, pharmacokinetics, and interactions with heparan sulfate-binding growth factors differ, as do their potential HPSE-independent effects.

Ferro et al. [4] showed that incorporating lipophilic aglycone functionality into sulfated oligosaccharide glycosides could improve pharmacological features, including preliminary intravenous half-life, while maintaining inhibition of heparanase and angiogenic growth-factor signaling. The resulting PG545 series illustrates how deliberate molecular design can improve exposure properties even within a sulfated glycan scaffold. Dredge et al. [5] subsequently documented antitumor and antimetastatic activity in preclinical models; Ritchie et al. [3] showed that SST0001 interfered with heparanase/syndecan-1 biology and reduced myeloma growth and angiogenesis. Such results supply genuine disease-model pharmacology, although polypharmacology complicates assignment of the total antitumor benefit to HPSE catalysis alone.

3.1.2 Defined glycans, sulfation pattern, and multivalency

Structurally defined glycomimetics offer a distinct optimization strategy in which sulfation topology, molecular architecture, and multivalent presentation can be varied. Zubkova et al. [30] reported discrete dendrimer glycomimetics with low-nanomolar enzyme inhibition, reduced tumor burden in a myeloma xenograft model, and favorable cytotoxicity and anticoagulant observations under the tested conditions. Loka et al. [31] systematically varied glycopolymer sulfation, measured binding to other heparan sulfate-binding proteins, and

demonstrated antimetastatic activity in mice. These studies link chain presentation and sulfation topology to biochemical activity and selected biological effects.

Shi et al. [32] investigated sulfated hyaluronan, reporting HPSE binding, reduced enzymatic activity, inhibition of carcinoma-cell invasion, and modulation of chemokine-associated signaling. Wakpal et al. [33] used aminoglycosides as accessible starting materials for defined heparan sulfate-like oligosaccharides and examined HPSE interaction relative to platelet factor 4. Hotor et al. [34] varied sulfation and hydrophobic capping groups in tobramycin-derived pseudo-trisaccharides. In their matched series, N-sulfated compounds generally inhibited more strongly than O-sulfated analogues; selected N-sulfated compounds reached submicromolar activity in a TR-FRET assay using biotinylated heparan sulfate as substrate. Hydrophobicity alone did not predict activity. Their greater than 300-fold selectivity result applies to selected protein-binding comparisons, not a comprehensive safety panel. Abdulsalam et al. [35] further modified paromomycin- and neomycin-derived glycans with sulfation and hydrophobic groups and reported tumor-model activity, extending the chemistry into in-vivo testing without establishing a matched exposure–target-engagement relationship. Ni et al. [49] found that synthetic glycol-split heparin trisaccharides were more potent than intact uronic-acid analogues, with STD-NMR and docking results aligning with enzyme-assay findings. Sugimoto et al. [50] reported a synthetic fucoidan analogue with approximately threefold greater HPSE inhibitory activity than its comparator and no measured factor Xa inhibition; it also bound SARS-CoV-2 spike and inhibited viral infection in cell assays. Because the antiviral mechanism can involve spike binding as well as HPSE inhibition, those results do not by themselves establish HPSE-dependent cellular target engagement. Across these glycan series, defined structural changes improve interpretability, but selected counterscreens and cell-line results do not establish comprehensive safety or drug disposition.

Wang et al. [45] subsequently applied 3D-QSAR to this aminoglycoside-derived series and profiled lead L17 in HPSE-dependent cancer cell lines. They reported concentration-dependent effects on proliferation and invasion, reduced extracellular HPSE levels, and minimal CYP3A4 inhibition in an experimental assay despite an in-silico liability flag. This extends the evidence from enzyme inhibition toward cellular activity and a limited metabolic counter-screen, but does not establish in-vivo exposure, systemic safety, or efficacy.

Recent polymeric mimetics challenge the presumption that molecular miniaturization is always preferred. Singh et al. [36] reported myeloma-model activity for an HPSE-inhibiting polymeric mimetic. Teki et al. [38] examined sulfated, sulfonated, and multivalent thioglycoside designs. Wang et al. [37] synthesized defined heparan sulfate oligosaccharides, probed substrate specificity, and reported a submicromolar inhibitor. Together, these studies examine charge placement, structural definition, multivalency, and substrate mimicry as independent design variables. Their biochemical activity remains distinct from established systemic bioavailability or clinical benefit.

3.2 Small-molecule chemotypes and experimentally supported SAR

3.2.1 Early heterocyclic inhibitor series

The synthetic small-molecule field predates modern HPSE structural biology. Courtney et al. [11, 12] disclosed isoindole and heteroaryl acetic-acid derivatives, establishing medicinal chemistry starting points without requiring large sulfated carbohydrate scaffolds. Rondanin et al. [14] investigated arylamidonaphthalene sulfonates inspired partly by antimetastatic suramin-related chemistry and described inhibitory activity

alongside selected cellular effects. These reports established chemical feasibility but did not provide the uniform selectivity, exposure, and target-engagement datasets now expected in a robust lead-optimization campaign.

Messore et al. [15] provide unusually interpretable within-series SAR for symmetrical benzazolyl inhibitors measured with a fondaparinux-based HPSE assay. In the nonfluorinated pair, replacing the urea linker in 5a (IC_{50} = 3.20 μM) with thiourea in 5d improved potency to 1.81 μM. The same linker change had the opposite effect in the fluorinated pair: urea-containing 5c inhibited at 0.18 μM, whereas thiourea 5e inhibited at 1.40 μM. Fluorination itself improved the urea analogue from 5a to 5c by about 18-fold. Extending the terminal carboxymethyl groups of 5e to glycine amides in 7g was associated with a further 17.5-fold potency increase (1.40 to 0.08 μM). Thus, the effect of linker identity depends on the surrounding substitution pattern, while the glycine extension is a promising within-assay matched-pair result rather than a general rule about polarity or charge. Figure 1 displays the corresponding 2D structure depictions and same-assay activity values. Compound 7g was less potent in that assay than the roneparstat comparator (0.005 μM). Selected compounds also reduced invasion and altered proangiogenic gene-expression readouts, but those phenotypes do not by themselves establish cellular HPSE target engagement or clinical exposure.

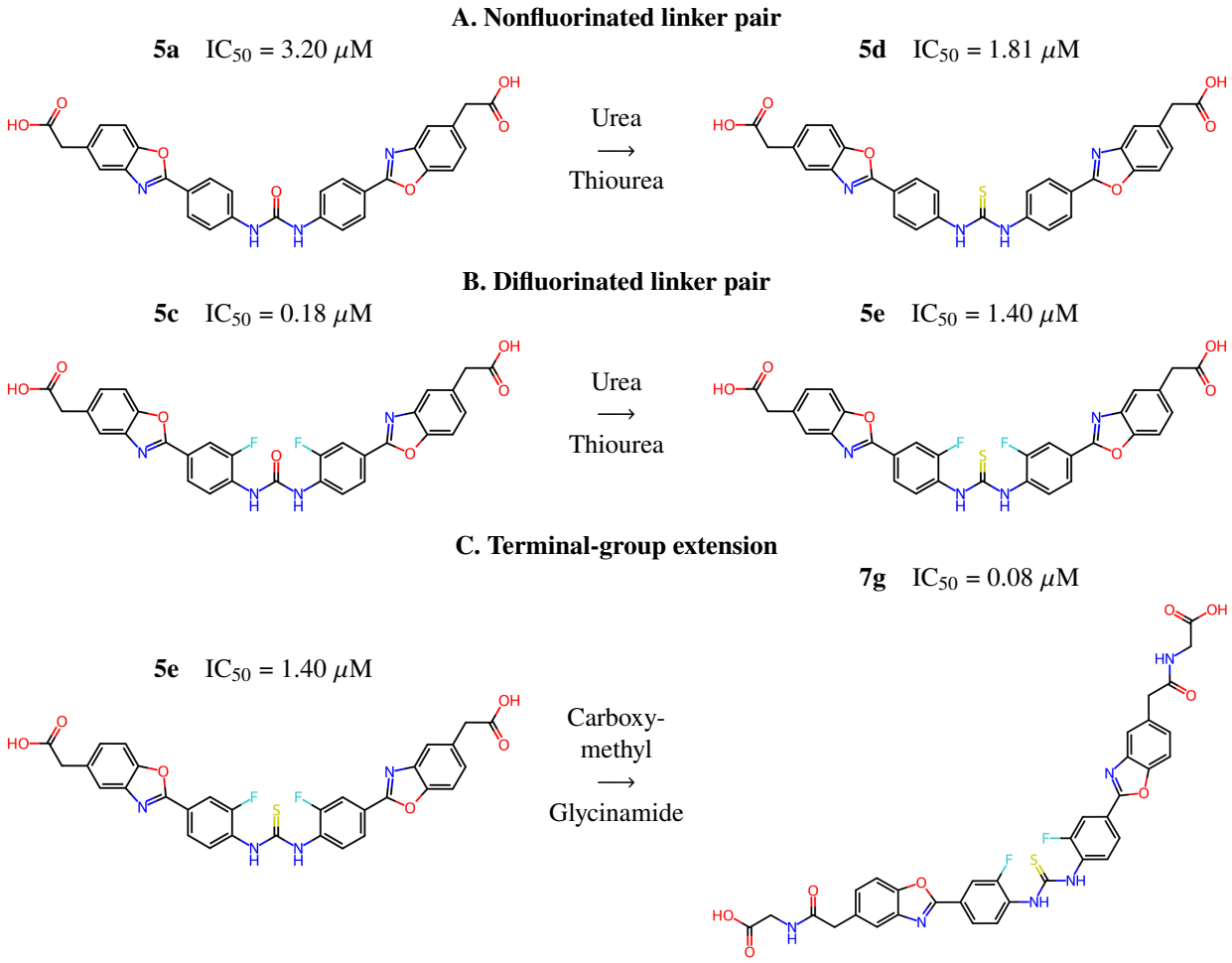


Figure 1: Matched-pair structure–activity relationships of symmetrical benzazolyl HPSE inhibitors reported by Messore et al. [15]. Urea-to-thiourea substitution produces opposite potency trends in nonfluorinated (A) and difluorinated (B) analogues. Terminal glycinamide extension of 5e yields the more potent analogue 7g (C). IC_{50} values were determined using the reported fondaparinux-based assay.

Pala et al. [16] tested 55 compounds selected through iterative ligand-based virtual screening; 19 showed IC_{50} values below 5 μM in the reported assay. Terephthalamide and isophthalamide scaffolds thus add experimentally active chemotypes beyond the established heterocyclic series, although their selectivity and pharmacological properties remain less characterized.

3.2.2 Triazolo-thiadiazoles and biological efficacy

Baburajeev et al. [17] screened 150 molecules from multiple scaffold families and identified triazolo-thiadiazole compounds with HPSE inhibitory activity. Their testing used radiolabeled extracellular-matrix substrate and cell-invasion experiments. Barash et al. [18] reported suppression of primary tumor growth and metastasis in experimental models. Together, the biochemical, cellular, and animal findings extend evidence for triazolo-thiadiazoles beyond purified-enzyme inhibition; the contribution of HPSE inhibition to tumor suppression remains unresolved because exposure-linked pharmacodynamics and causal mechanistic controls are limited.

3.2.3 Tetrahydroimidazopyridines: explicit attention to selectivity

The tetrahydroimidazo[1,2-a]pyridine-5-carboxylic acid program reported by Imai and colleagues illustrates a different matched-pair question from the benzazolyl series. X-ray-guided and fragment-molecular-orbital-supported optimization produced compound 4e, with more than sevenfold improved HPSE1 inhibition over the initial lead 2. Subsequent methylation of the 6-hydroxyl group yielded compound 18: HPSE1 activity was retained while inhibition of exo-β-D-glucuronidase and glucocerebrosidase was reduced. Compound 18 also significantly suppressed mouse serum HPSE1 activity after dosing at 3, 30, and 100 mg/kg, with the effect lasting up to 4 h [19–21]. The 4e-to-18 change therefore improved enzyme selectivity while retaining HPSE1 activity and systemic pharmacodynamics; diseased-tissue engagement and human pharmacokinetics or efficacy remain unreported. A companion investigation shortened a demanding synthesis by five steps for selected analogues.

3.2.4 Mechanism-based cyclophellitols and structurally characterized inhibitors

De Boer et al. [22] appended an HS-like α-1,4-GlcNAc unit to a glucurono-cyclophellitol epoxide, producing pseudodisaccharide inhibitors with nanomolar potency and irreversible HPSE inhibition; they also reported reduced cancer-associated invasion and murine metastasis. Replacing GlcNAc with glucosamine abolished measurable HPSE inhibition up to 100 μM, implicating the recognition element in activity beyond the covalent warhead. Borlandelli et al. [23] varied 4-O-substitution; selected alkyl analogues retained low-micromolar HPSE inhibition and improved selectivity over exo-β-D-glucuronidase, supported by structural and computational analysis. Thus, glycan recognition and 4-O-substitution affect distinct properties—potency and enzyme selectivity, respectively. Ehren et al. [24] further expanded functionalized cyclophellitol chemistry. Selectivity among glycosidases, off-target covalent labeling, recovery duration, and route suitability remain open questions.

Chen et al. [27] showed crystallographically and kinetically that siastatin B forms distinct inhibitory chemical

species, with activity depending on enzyme and binding context. Its mechanism therefore cannot be inferred from the starting structure alone.

3.2.5 Fragment binding and emerging chemical space

Of the 31 structurally observed fragment binders reported by Davies et al. [25], five inhibited HPSE at micromolar concentrations; fragment growth improved potency approximately sevenfold. These crystallographically supported orthosteric and allosteric binders remain early-stage findings, with cellular engagement, exposure, and in-vivo pharmacology unreported. The representative series in Table 1 show the range of evidence available across chemotypes.

Table 1: Selected inhibitor series with experimental evidence. Potencies are study-specific and not ranked across assays. "NR in abstract" means the indexed abstract does not specify substrate/readout; it does not imply omission from the full text. "Not reproduced in this review" marks a value omitted from this table. NR in the final column denotes an evidence domain not established in the primary source examined.

Study and chemotype	**Measured biochemical evidence (assay/substrate)**	**Other measured evidence**	**Evidence gap (not established in source)**
Courtney et al. [11, 12], heterocyclic acids	Enzyme inhibition across synthesized analogues; assay/substrate varies by report	Early SAR	Consistent counterscreens, cellular target engagement, exposure, and in-vivo efficacy NR
Messore et al. [15], symmetrical benzazolyl series	5a 3.20; 5d 1.81; 5c 0.18; 5e 1.40; 7g 0.08 μM; fondaparinux-based HPSE assay	Invasion and proangiogenic readouts for selected compounds	HPSE-dependent cellular effects and systemic exposure NR
Pala et al. [16], virtual-screening hits	19/55 compounds with $IC_{50} < 5$ μM; human HPSE assay with fondaparinux substrate and WST-1 readout	Cellular invasion and gene-expression tests on selected hits	Selectivity, systemic exposure, and in-vivo efficacy NR
Baburajeev et al. [17], triazolo-thiadiazoles	Direct HPSE activity screening with radiolabeled extracellular-matrix substrate	Invasion experiments	Cellular HPSE dependence and exposure-linked pharmacology NR
Barash et al. [18], triazolo-thiadiazoles	HPSE inhibitory activity; quantitative potency not reproduced in this review	Tumor and metastasis experiments in animals	Target-site exposure and HPSE-specific causal efficacy NR
Imai et al. [19–21], tetrahydroimidazopyridines	Structure-guided potency optimization; 4e-to-18 matched methylation retained HPSE1 activity	Selectivity over GUSβ/GBA; serum HPSE1 pharmacodynamics for 18	Human PK, diseased-tissue engagement, and efficacy NR
De Boer et al. [22], cyclophellitol derivatives	Nanomolar irreversible HPSE inhibition; GlcNAc-to-glucosamine substitution abolished activity up to 100 μM	Cellular and mouse metastasis outcomes	Long-term off-target selectivity and human exposure NR

Study and chemotype	Measured biochemical evidence (assay/substrate)	Other measured evidence	Evidence gap (not established in source)
Borlandelli et al. [23], substituted cyclophellitols	4-O-substituted analogues retain low-micromolar HPSE inhibition; substrate/readout NR in abstract	Exo-β-glucuronidase selectivity and crystallographic validation	Cellular/in-vivo efficacy and exposure NR
Zhang et al. [26], TC LPA5 4	IC_{50} 10 μM in HADP fluorogenic assay; MX4-62 experimentally inactive	Docking and MD, including computational analog designs	Prospective validation of proposed analog SAR and pharmacology NR
Davies et al. [25], fragment series	Five of 31 binding fragments inhibited in the authors' enzyme assay; micromolar activity and approximately sevenfold gain after fragment growth; substrate/readout NR in abstract	Experimental fragment binding structures	Cellular target engagement, exposure, and in-vivo efficacy NR
Zubkova et al. [30], dendrimer mimetics	Low-nanomolar inhibition	Anticoagulant/cytotoxicity tests and myeloma model	Systemic exposure, target-site engagement, and clinical efficacy NR
Loka et al. [31], defined glycopolymers	Potent enzyme inhibition	Competition against HS-binding proteins; mouse antimetastatic activity	Human evidence and exposure-linked target engagement NR
Hotor et al. [34], tobramycin glycans	Matched sulfation/hydrophobicity series; TR-FRET assay with biotinylated HS substrate	Comparative PF4/antithrombin binding; cell-line activity	Broad proteome selectivity, systemic exposure, and in-vivo efficacy NR
Wang et al. [37], defined HS oligosaccharides	HPSE inhibition assay: inhibitory nonasaccharide IC_{50} 0.761 μM; substrate/readout NR in abstract	Substrate-cleavage mechanism studies	Cellular target engagement, exposure, and clinical safety NR
Wang et al. [45], aminoglycoside glycan L17	3D-QSAR-guided follow-up of sulfated and hydrophobic analogues	Cancer-cell proliferation, invasion, extracellular HPSE, CYP3A4 counter-screen	In-vivo exposure, systemic safety, and efficacy NR
Ni et al. [49], glycol-split heparin oligosaccharides	Glycol-split trisaccharides more potent than intact analogues	STD-NMR binding data aligned with docking and activity	In-vivo exposure, target engagement, and safety NR
Sugimoto et al. [50], synthetic fucoidan analogue	Approximately threefold greater HPSE inhibition than comparator	Factor Xa counter-screen; SARS-CoV-2 spike-binding and cell-infection assays	HPSE-dependent antiviral mechanism and in-vivo pharmacology NR
Zhou et al. [40], sulfated tetrasaccharides	HPSE inhibition assay: compound 5 IC_{50} 11.27 μM; substrate/readout NR in abstract	Sepsis survival, inflammatory and morphological outcomes in mice	Target-site exposure and HPSE-specific causal contribution NR
Mo et al. [51], low-molecular-weight dextran sulfate	Direct HPSE binding and inhibition; reduced HPSE expression	Invasion, angiogenesis, mouse metastasis and safety assessments	Exposure and HPSE-specific in-vivo causality NR
Crompton et al. [52], OVZ/16-35 glycomimetic	Tetravalent HS mimic	Rat MCD model; glycocalyx protection and reduced albuminuria	Human PK, bioavailability, and long-term safety NR

3.3 Why biochemical potency cannot be compared without assay context

3.3.1 Distinct substrates yield distinct measurements

Published assays use natural or radiolabeled extracellular matrix heparan sulfate, heparin-derived fragments, fondaparinux, fluorescence-generating synthetic substrates, and other reporting systems. These methods differ in substrate size, sulfation, matrix presentation, readout physics, turnover conditions, and potential compound interference. A compound may act as a competitive substrate-site blocker under one condition but display apparent mixed inhibition or reduced effectiveness under another. Liu et al. [29] introduced an ultrasensitive fluorogenic probe that greatly improved practical activity detection, whereas Whitefield et al. [28] demonstrated how glycomimetic inhibition can depend on ligand size and multiple binding modes.

Published IC_{50} values depend on enzyme preparation, substrate concentration, detection system, incubation time, and data fitting; they are not universal affinity constants. A low-nanomolar value for a multivalent glycopolymer cannot be directly ranked against a small molecule measured on a different substrate and platform. Table 1 identifies assay and substrate where reported.

Wang et al. [47] developed a single-heparan-sulfate-substrate isotope-dilution mass-spectrometry method (SHS-IDMS) for quantifying HPSE activity in plasma. Quantification of products from a defined 12-mer by LC-MS/MS distinguished samples from children with sepsis and healthy children; activity also correlated with syndecan-1 and angiopoietin-2. The method supports measurement in complex biological matrices, but has not been validated as an inhibitor-response assay across chemotypes or dosing studies. The authors disclosed commercial interests related to glycan therapeutics.

3.3.2 Orthogonal confirmation and interference controls

Fluorescence and colorimetric hits can reflect optical interference, aggregation, nonspecific polyanion interactions, or reporter effects, particularly for charged glycomimetics and chromophoric compounds. Orthogonal assays and counterscreens against related glycosidases and relevant heparan sulfate-binding proteins strengthen inhibition and selectivity claims. SPR, NMR, thermal-shift, or crystallographic binding alone does not establish catalytic inhibition. Where primary reports do not specify relevant controls, these are recorded as not reported.

3.4 Selectivity, structural evidence, and target engagement

HPSE selectivity spans enzyme-family discrimination, binding to other heparan sulfate-associated proteins, and cellular effects involving catalytic or noncatalytic HPSE functions. Imai et al. [19–21] provide direct evidence of enzymatic selectivity optimization, while Borlandelli et al. [23] combine selectivity measurements with structure-guided analysis. Loka et al. [31], Wakpal et al. [33], and Hotor et al. [34] show that selected interactions of sulfated ligands can be modified through structure. These panels characterize chosen targets, not comprehensive pharmacological specificity.

In Messore et al. [15], benzazolyl compounds inhibited purified HPSE and reduced invasion and proan-

giogenic readouts, but these cellular phenotypes do not establish HPSE dependence. Imai et al. [19–21] reported post-dose suppression of mouse serum HPSE activity, a proximal pharmacodynamic measure that does not establish engagement in diseased tissue. Loka et al. [31] linked defined sulfation patterns and selected protein-binding comparisons to antimetastatic activity in mice, but the animal outcome is not a direct occupancy measure.

4 Translational evidence from models to clinic

4.1 Cellular and preclinical activity: compelling but heterogeneous evidence

Cellular or animal observations accompany biochemical evidence for benzazolyl inhibitors [15], triazolothiadiazoles [17, 18], and cyclophellitols [22]. Differences in tumor models, dosing schedules, invasion assays, and compound exposure preclude direct comparison.

Among newer glycomimetic studies, Mo et al. [51] reported that low-molecular-weight dextran sulfate sodium bound HPSE, inhibited its activity and expression, reduced 4T1-cell invasion and endothelial angiogenesis, and suppressed pulmonary metastasis in mice. The accompanying safety assessment strengthens its preclinical profile; exposure and target-specificity characterization remain limited. Such polymeric agents have reached animal efficacy studies, whereas newer small-molecule fragments remain at an earlier stage.

Nononcological models include diabetic microvascular injury [41], sepsis [39, 40], pulmonary fibrosis [42], chronic obstructive pulmonary disease [43], and minimal change disease [52]. In sepsis, Zhou et al. [40] reported that sulfated tetrasaccharide 5 inhibited HPSE (IC_{50} = 11.27 μM) and improved survival and inflammatory outcomes in mice. In a rat model of minimal change disease, OVZ/16-35 protected the endothelial glycocalyx and reduced albuminuria [52]; pharmacokinetics, bioavailability, and long-term safety remain under study, and the authors disclosed related patents. Variation in disease mechanism, timing, route, and endpoint complicates comparison across these models.

Few animal studies pair efficacy with target-tissue exposure and proximal HPSE pharmacodynamics, limiting interpretation of whether the compound reached the target or acted through HPSE alone.

4.2 Human pharmacology and clinical outcomes

Human studies provide the most consequential check on preclinical assumptions. Basche et al. [6] examined PI-88 in advanced solid tumors, establishing an early clinical pharmacology and tolerability foundation. Hudachek et al. [7] later developed a population pharmacokinetic model for PI-88. A randomized Phase II dose-finding study after curative HCC resection enrolled 172 patients (168 in the intention-to-treat population); at 48 weeks, recurrence-free proportions were 50% with placebo, 63% with 160 mg PI-88, and 41% with 250 mg. The study supported selection of the 160-mg dose but was not a confirmatory efficacy trial, and the non-monotonic dose pattern warrants caution [53]. In the subsequent Phase III PATRON trial, 520 patients were randomized and 519 were included in the intention-to-treat analysis. The primary disease-free-survival endpoint was not met. A favorable result reported for the per-protocol subgroup with microvascular invasion is exploratory and should not override the negative overall primary result; these findings were

reported in a conference abstract [54]. A Phase I roneparstat investigation in multiple myeloma reported tolerability within the studied dose levels, but early safety observations did not demonstrate established clinical efficacy.

The Phase I pixatimod monotherapy study enrolled 23 patients with advanced solid malignancies. Dose escalation identified 100 mg weekly as the maximum tolerated dose; no objective responses were observed [9]. A subsequent Phase Ib study combined pixatimod with nivolumab and reported three partial responses among 25 evaluable patients with microsatellite-stable metastatic colorectal cancer, but none in the metastatic pancreatic cancer cohort. The combination MTD was 25 mg pixatimod weekly with nivolumab; one treatment-related grade 5 multiorgan-failure event occurred at the 50-mg pixatimod dose [55]. Responses cannot be attributed to HPSE inhibition alone because nivolumab was co-administered, pixatimod has immunomodulatory activity, and the study did not isolate an HPSE-specific pharmacodynamic contribution.

In a randomized Phase II trial in untreated metastatic pancreatic ductal adenocarcinoma, necuparanib was added to gemcitabine and nab-paclitaxel. Median overall survival was 10.71 months with necuparanib and 9.99 months with placebo (hazard ratio, 1.12; 95% CI, 0.66–1.89; $P = 0.671$), with the point estimate favoring placebo. Median progression-free survival was 5.52 versus 6.93 months, respectively (hazard ratio, 0.97; 95% CI, 0.61–1.54; $P = 0.886$), and response rates were comparable. Hematologic toxicity was reported in 83% of treated patients versus 70% with placebo. The report is internally inconsistent about enrollment: its abstract says 110 patients were randomized, whereas the Results section reports 120 (62 assigned necuparanib and 58 placebo), consistent with the arm counts. The study crossed a planned futility boundary and showed no overall-survival benefit [10].

Chen et al. [48] later reported that PI-88 suppressed stromal activation in coculture and improved tumor control and survival with gemcitabine and nab-paclitaxel in orthotopic PDAC models. The study offers a preclinical combination rationale, but concerns PI-88 rather than necuparanib and reports the underlying data as confidential; it does not establish clinical benefit.

Taken together, PI-88's missed Phase III primary endpoint, necuparanib's negative randomized Phase II result, and pixatimod's combination responses establish mixed clinical evidence; none isolates an HPSE-specific pharmacodynamic contribution. These studies characterize human administration and tolerability but do not link target-site HPSE suppression to exposure and outcome. Table 2 summarizes the clinical evidence for PI-88, roneparstat, pixatimod, and necuparanib.

Table 2: Clinical studies of heparan sulfate mimetics. HCC, hepatocellular carcinoma; ITT, intention-to-treat; DFS, disease-free survival; MVI, microvascular invasion; MSS, microsatellite-stable; mCRC, metastatic colorectal cancer; mPDAC, metastatic pancreatic ductal adenocarcinoma; OS, overall survival; PFS, progression-free survival.

Program	**Phase and setting**	**Reported evidence**	**Interpretation for lead development**
PI-88 / muparfostat	Phase I advanced tumors [6]; Phase II post-resection HCC [53]	Phase II: 172 randomized, 168 ITT; 48-week recurrence-free proportions 50% placebo, 63% at 160 mg, and 41% at 250 mg	Dose-finding study supported 160 mg for further study; recurrence findings were preliminary, not confirmatory efficacy

Program	Phase and setting	Reported evidence	Interpretation for lead development
PI-88 / muparfostat (PATRON)	Phase III adjuvant HCC after resection; conference abstract [54]	520 randomized, 519 ITT; primary DFS endpoint not met; favorable per-protocol MVI subgroup result	Overall result negative; MVI subgroup exploratory
Roneparstat / SST0001	Phase I, multiple myeloma [8]	Acceptable tolerability in tested dose cohorts	Early safety does not demonstrate efficacy
Pixatimod / PG545 monotherapy	Phase I, 23 advanced solid-tumor patients [9]	MTD 100 mg weekly in the studied regimen; no objective responses	Human exposure/tolerability characterized; no monotherapy objective response
Pixatimod / PG545 plus nivolumab	Phase Ib, advanced solid tumors; MSS mCRC and mPDAC cohorts [55]	Combination MTD: pixatimod 25 mg weekly; 3 partial responses among 25 evaluable MSS mCRC patients, none in mPDAC; treatment-related grade 5 multiorgan failure at 50 mg pixatimod	Responses cannot be assigned to HPSE inhibition; high-dose toxicity noted
Necuparanib / M-402	Randomized Phase II metastatic pancreatic cancer [10]	OS 10.71 vs 9.99 months (HR 1.12); PFS 5.52 vs 6.93 months (HR 0.97); comparable response rates; hematologic toxicity 83% vs 70%. Abstract reports 110 randomized; Results report 120 (62 vs 58)	No OS/PFS benefit; added hematologic toxicity without demonstrated efficacy gain; enrollment total internally inconsistent

5 Computational discovery and emerging modalities

5.1 Computational discovery: hypothesis generation versus validation

Gozalbes et al. [13] combined structure- and ligand-based hit identification with experimental assays.

Shanbhogue et al. [46] trained an HPSE activity classifier on 225 curated ChEMBL measurements, reporting 80% overall and 78.5% balanced accuracy on a stratified 20% holdout set. They predicted activity for 14 external compounds and ranked 900 screening candidates, but reported no biochemical testing of these proposed hits. The library screen used 2D descriptors and Morgan fingerprints, whereas the selected model included an additional 3D descriptor; the feature-space difference is not explained. Zhang et al. [26] likewise illustrate the gap between prediction and validation: TC LPA5 4 inhibited HPSE (IC_{50} = 10 μM), but its synthesized analog MX4-62 was inactive despite docking and molecular-dynamics rationale. Thus, the reported MX4-62 inactivity is experimental, while the proposed SAR for further analogues remains computational. Together, these studies show that computational prioritization has not yet consistently yielded experimentally confirmed HPSE inhibitors.

5.2 Emerging HPSE-targeting modalities and boundaries of this review

Li et al. [44] reported JW-9, a heparanase-directed lysosome-targeting chimera (LYTAC) designed to promote extracellular HPSE degradation and improve natural killer cell antitumor effects in experimental models. Unlike catalytic inhibitors, JW-9 targets extracellular protein abundance, making it a distinct therapeutic modality rather than a conventional small-molecule active-site inhibitor.

Agents that modify heparan sulfate availability or growth-factor recognition without demonstrating direct HPSE activity provide biological context but are not classified here as biochemical inhibitors.

6 Lead appraisal and future directions

6.1 Evidence-based appraisal of lead quality

Biochemical hits, mechanistically supported inhibitors, biologically validated leads, and pharmacologically characterized candidates represent distinct evidence levels, not fixed IC_{50} thresholds. Potency alone cannot qualify an unselective or poorly characterized compound, while a micromolar inhibitor may remain a productive lead when its mechanism, selectivity, formulation, exposure, and cellular response are supported.

Confidence increases when substrate-defined inhibition is confirmed orthogonally and supported by modality-appropriate selectivity, target engagement, exposure, safety, and disease-relevant outcomes. Highly sulfated glycans, covalent inhibitors, degraders, and conventional small molecules require different selectivity and engagement measurements.

Figure 2 maps evidence domains for candidate-specific claims; Figure 1 illustrates context-dependent matched-pair SAR.

Evidence remains uneven across modalities: Messore's 7g links matched-pair potency to invasion phenotypes; the Imai series adds enzyme selectivity and serum pharmacodynamics; de Boer et al. report mouse outcomes for cyclophellitols; clinical glycomimetics provide human exposure data but mixed efficacy. No single example establishes the full path from biochemical activity to therapeutic benefit. Table 3 summarizes the scope and limits of each evidence domain.

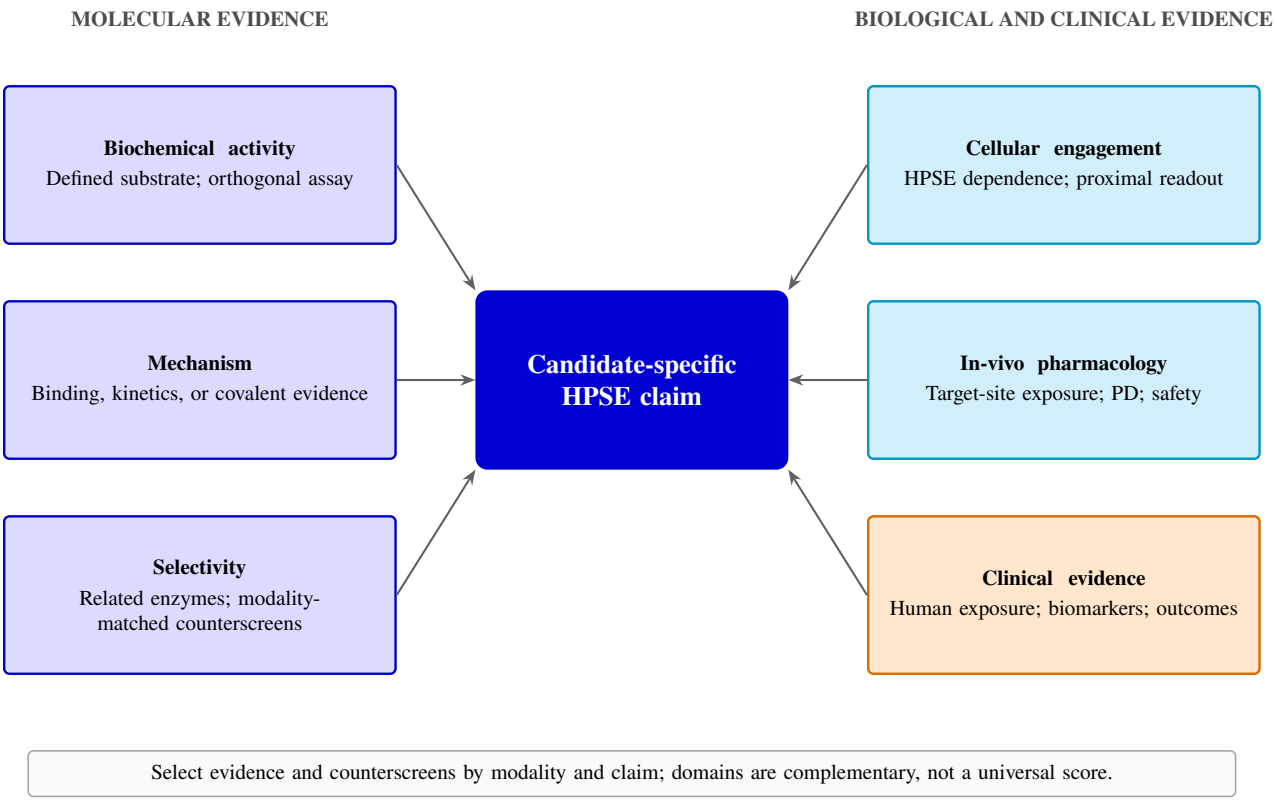


Figure 2: Complementary evidence domains for evaluating HPSE-directed candidates. Molecular, cellular, in-vivo, and clinical measurements support claims about mechanism, selectivity, target engagement, and therapeutic relevance.

Table 3: Experimental evidence domains and interpretive limits in HPSE inhibitor evaluation.

Evidence domain	**Typical measurements**	**What it establishes**	**What it does not establish alone**
Biochemical inhibition	IC_{50}, K_i, inhibition kinetics under stated substrate conditions	Effect on a defined enzyme assay	Whole-organism potency or mechanism specificity
Assay confirmation	Orthogonal enzyme assay, interference control, repeatability	Greater confidence that the biochemical effect is genuine	Cellular target engagement
Selectivity	GUSβ/GBA counterscreens; PF4, antithrombin, growth-factor binding panels	Reduced activity at chosen alternative targets	Complete proteome-wide selectivity or safety
Structural mechanism	Cocrystal structure, kinetic model, covalent labeling, biophysical binding	More direct explanation of enzyme–ligand interactions	Effective dose or disease benefit
Cell-based relevance	HPSE activity in cells, invasion, matrix degradation, genetic/rescue controls	Relevant cellular pharmacology, strongest with mechanistic controls	Systemic exposure or clinical efficacy
Animal validation	PK/PD relationship, tumor growth, metastasis, inflammation, survival	In-vivo proof of concept in a specified model	Efficacy in humans
Clinical development	Exposure, tolerability, response rates, randomized outcomes	Human safety and/or efficacy under tested regimen	Transferability to other indications

6.2 Research gaps and future directions

Assay comparability: A common reference panel spanning distinct chemotypes, tested in orthogonal assay formats using distinct substrates or readout principles, would clarify scaffold-dependent assay behavior. Reports should specify enzyme construct and active fraction, substrate and kinetic regime, interference controls, and curve-fitting uncertainty.

Mechanistic specificity: Cellular studies need to separate catalytic from noncatalytic HPSE effects and distinguish HPSE inhibition from growth-factor sequestration or general matrix disruption. Genetic or rescue controls can support causal attribution; glycomimetics warrant binding panels, while covalent inhibitors require proteome-reactivity and enzyme-family selectivity assessment.

Medicinal chemistry: Matched-pair synthesis and measurement of solubility, protein binding, formulation, and stability can strengthen SAR before extensive animal studies. Computational SAR remains predictive until experimentally tested; the inactivity of MX4-62 [26] illustrates the value of reporting negative outcomes.

Translation: Animal efficacy paired with target-tissue exposure and proximal HPSE pharmacodynamics would clarify exposure-response relationships. Indication selection should account for measurable HPSE dependence, intervention timing, route, and accessible biomarkers; oncology, sepsis, glycocalyx dysfunction, and fibrosis pose distinct therapeutic contexts. Table 4 summarizes reporting priorities.

Table 4: Recommended reporting elements for HPSE inhibitor studies.

Reporting package	Study-specific information to report
Enzyme assay	Enzyme construct and maturation state; substrate identity and concentration relative to kinetic conditions; incubation and readout; replicate number, raw dose-response data, and fitting uncertainty
Assay confirmation	Orthogonal detection chemistry; optical, aggregation, and reporter-interference controls; matched positive and negative compounds
Selectivity panel	Comparator enzymes or heparan sulfate-binding proteins selected for the chemotype, with concentrations and assay conditions
Mechanism	Kinetic model and evidence for reversible, covalent, allosteric, or abundance-modulating action; distinguish measured binding from predicted poses
Cellular studies	Cell exposure and viability; proximal HPSE activity or abundance readout; genetic, rescue, or structurally distinct inhibitor controls where available
In-vivo studies	Dose, route, formulation, time-matched plasma and target-tissue exposure, pharmacodynamic readout, safety measures, and efficacy endpoints
Clinical studies	Dose and exposure; prespecified target-proximal biomarker and sampling time; eligibility criteria, treatment context, and clinical endpoints

7 Limitations of the evidence synthesis

As a narrative review based on focused searches, this synthesis may not include every relevant study. Representative studies were selected to illustrate distinct chemical and translational evidence profiles, not to estimate the relative frequency or overall success of each inhibitor class. Variation in assay methods and reporting limits comparisons of potency, target engagement, and animal outcomes; some mechanistic interpretations also depend on experimental details unavailable in the examined sources. The conclusions therefore represent a critical interpretation of selected literature, not an exhaustive ranking of HPSE inhibitor classes.

8 Conclusions

Matched-pair SAR reveals context-dependent design constraints: urea-to-thiourea substitution reverses its potency effect across benzazolyl substitution patterns; 4e-to-18 methylation improves enzyme selectivity while retaining HPSE1 activity and serum pharmacodynamics; cyclophellitol glycan changes affect recognition and selectivity; and aminoglycoside activity reflects both sulfation and hydrophobic capping. Clinical findings remain mixed: PI-88 missed the PATRON primary endpoint, necuparanib showed no survival benefit with greater hematologic toxicity, and pixatimod combination responses cannot be attributed to HPSE inhibition alone. Across chemotypes, the central limitation is the incomplete connection between biochemical activity, disease-relevant target engagement, exposure, and therapeutic benefit.

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